Will Rosenberg

Arizona State University
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Education

2025 – Present
Arizona State University
PhD Student in Computer Science
Advisor: Yan Shoshitaishvili

2021 – 2025
Washington University in St. Louis (WashU)
Bachelor of Science in Computer Science, Physics, and Mathematical Sciences
Summa Cum Laude

Research Experience

Aug 2025 – Present
SEFCOM Laboratory
Arizona State University
- Exploring co-obfuscation techniques to prevent bug attribution.
- Developing emulation and fuzzing techniques to detect bugs in macOS kernel extensions.
- Implementing automated userspace program recording and fuzzing for IPC-based programs.

Aug 2023 – May 2025
Computer Security and Privacy Laboratory
Washington University in St. Louis
- Applied fuzzing techniques (libFuzzer, AFL) to identify potential vulnerabilities in open-source cyber-physical libraries written in C/C++.
- Surveyed and systematized timing bugs across open-source real-time operating systems including FreeRTOS, Zephyr, and RIOT OS.

May – Dec 2022
Quantum Monte Carlo Group for Nuclear Physics
Washington University in St. Louis
- Used a neural network to approximate the numerical solution of a nucleon-nucleon scattering problem with the goal of applying this network in the optimization problem of nuclear interaction models.

Technical Experience

2025 – 2026
Open-Source Contributions
- Discovered and patched a Protobuf vulnerability, earning **$3,133.70** from Google's OSS VRP.
- Authored upstream Linux kernel fixes for **CVE-2025-71085**, **CVE-2026-23127**, and multiple other bugs; independently discovered and reported **CVE-2026-23014** before public disclosure, then developed a local privilege escalation exploit and technical write-up.
- Extended Juvenal, an AI-agent orchestrator, to support dynamic workflows and verification loops for vulnerability discovery in open-source software.

Aug 2022 – Present
Shellphish & B34R5HELL: Capture-the-Flag (CTF) Competitions
- Compete in 48-hour CTF hacking competitions with a focus on binary exploitation and experience in reverse engineering and cryptography.
- Qualified for and competed in DEF CON CTF 2026 in-person finals, m0leCon CTF 2026 finals, and Securinets 2025 finals with Shellphish.
- Captained B34R5HELL (2023 – 2025). Created an introductory bootcamp, prepared lectures and activities for weekly meetings, and taught topics in binary exploitation, cryptography, web security, reverse engineering, and forensics.

Achievements & Awards

2025 **Outstanding Senior Award**. WashU Computer Science Department
2025 **Departmental Distinction**. WashU Department of Physics
2025 **Sigma Pi Sigma**. National Physics and Astronomy Honor Society

Academic Service

2026 **USENIX WOOT 2026 Artifact Evaluation Committee Member**. Reviewed and tested research artifacts.
2025 **ACSAC 2025 CTF Organizing Committee Member**. Created a challenge exploring UAF in jemalloc.

Teaching

Arizona State University

2026 **Graduate Teaching Assistant**. Introduction to Cybersecurity

Washington University in St. Louis

2024 – 2025 **Head Teaching Assistant**. Systems Software
2021 – 2023 **Teaching Assistant**. Introduction to Computer Security; Object-Oriented Programming Workshop; Introduction to Computer Science

Publications & Presentations

Publications & Preprints

- Z. Zhang, A. P. Bajaj, D. Handa, S. Liu, A. S. Raj, H. Chen, H. Wang, Y. Liu, Z. L. Basque, S. Nath, **W. Rosenberg**, V. Juneja, N. Chapre, T. Bao, Y. Shoshitaishvili, A. Doupé, C. Baral, and R. Wang. "Build-Bench: Benchmarking LLM Agents on Compiling Real-World Open-Source Software". *Transactions on Machine Learning Research*, 2026. [paper]
- H. Wang, Z. L. Basque, J. Hu, A. P. Bajaj, Y. Liu, S. Zhu, G. Kobakhia, N. Chapre, **W. Rosenberg**, S. Mishra, A. M. Gabani, M. Schloegel, A. Doupé, Y. Shoshitaishvili, R. Wang, and T. Bao. "Root-Cause-Driven Automated Vulnerability Repair". *arXiv preprint*, 2026. [paper]
- **W. Rosenberg**, E. Herrera-Vendrell, R. Nucuta, K. Wahal, S. Bhardwaj, W. B. Gallie, and P. McClernon. "Predicting Orbital Resonance of 2867 Šteins Using the Yarkovsky Effect". *Journal of Emerging Investigators*, 2021. [paper]

Poster Presentations

- **W. Rosenberg**, M. Piarulli, A. Lovato, K. Raghavan, and S. Pastore. "Training a feed-forward neural network to solve the S-wave nucleon-nucleon scattering problem". Poster presented at the Midstates Consortium for Math and Science Undergraduate Research Symposium, Washington University in St. Louis, Nov. 2022. Also presented at the Washington University Physics Research Symposium.